

Class work problem unit 4

Problem Set 6.1

(a) What 3 by 3 symmetric matrices A_1 and A_2 correspond to f_1 and f_2 ?

$$f_1 = x_1^2 + x_2^2 + x_3^2 - 2x_1x_2 - 2x_1x_3 + 2x_2x_3$$

$$f_2 = x_1^2 + 2x_2^2 + 11x_3^2 - 2x_1x_2 - 2x_1x_3 - 4x_2x_3.$$

Q8: Check for positive definiteness of the following

$$(a) \begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix}, \quad (b) \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad (c) \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}, \quad (d) \begin{bmatrix} -1 & 2 \\ 2 & -8 \end{bmatrix}.$$

Q10:

8. If $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ is positive definite, test $A^{-1} = \begin{bmatrix} p & q \\ q & r \end{bmatrix}$ for positive definiteness.

Problem Set 6.2

Q1:

2. Decide for or against the positive definiteness of

$$A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}^2$$

A through pivot method, B through determinant method and C through eigen values.

Chapter 6.3

SVD

1. Find the Singular Value Decomposition (SVD) of the matrix

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$$

2. Find the Singular Value Decomposition (SVD) of the matrix

$$A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & 2 \end{bmatrix}.$$

3. Find the Singular Value Decomposition (SVD) of the matrix

$$A = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$$

PCA

1. Given the data matrix

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix},$$

perform Principal Component Analysis (PCA) using Singular Value Decomposition (SVD)

2. Given the dataset

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 2 \\ 3 & 1 \end{bmatrix},$$

perform Principal Component Analysis (PCA) using the covariance matrix method

Cholesky Decomposition

From the pivots, eigenvalues, and eigenvectors of $A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$, write A as $R^T R$ in three ways: $(L\sqrt{D})(\sqrt{D}L^T)$, $(Q\sqrt{\Lambda})(\sqrt{\Lambda}Q^T)$, and $(Q\sqrt{\Lambda}Q^T)(Q\sqrt{\Lambda}Q^T)$.